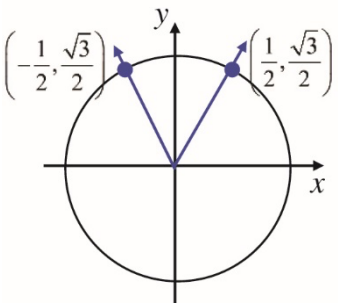
Solve Trigonometric Equations in Linear Form

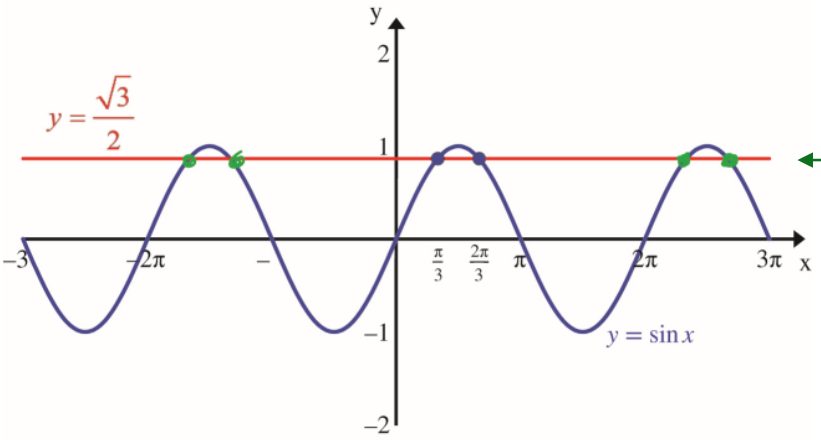
Example: Solve. $\sin x = \frac{\sqrt{3}}{2}$

From the unit circle, we can see that there are 2 values of x between 0 and 2π that will solve the equation: $x = \frac{\pi}{3}$ and $x = \frac{2\pi}{3}$.



However, there are actually infinitely many solutions to the equation:

$x = \frac{\pi}{3} + 2n\pi$ and $x = \frac{2\pi}{3} + 2n\pi$ (where n is any integer).



all intersections are solutions

---

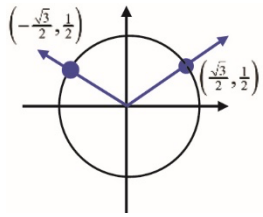
1. Solve the equation on the interval $[0, 2\pi)$. $3\csc x = 4 + \csc x$

$2\csc x = 4$

$\csc x = \frac{4}{2}$

$\sin x = \frac{1}{2}$

$x = \frac{\pi}{6}, \frac{5\pi}{6}$



2. Solve the equation over the set of real numbers. $\sqrt{2}\cos x + 5 = 4$

$\sqrt{2}\cos x = -1$

$\cos x = \frac{-1}{\sqrt{2}} = -\frac{\sqrt{2}}{\sqrt{2}}$

$x = \frac{3\pi}{4}, \frac{5\pi}{4}$

$x = \frac{3\pi}{4} + 2n\pi$ & $\frac{5\pi}{4} + 2n\pi$

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.

Solve Trigonometric Equations Involving Multiple or Compound Angles

Solve the equation on the interval $[0, 2\pi)$.

3. $8\cot 2x - 5 = 3$

Isolate $\cot 2x$.

Give the general solution from the unit circle.

To find solutions on the interval $[0, 2\pi)$, substitute $n = \ldots, -1, 0, 1, 2, \ldots$

$8\cot 2x = 8$

$\cot 2x = 1$

$2x = \frac{\pi}{4}, \frac{5\pi}{4}$

$2x = \frac{\pi}{4} + n\pi$ & $2x = \frac{5\pi}{4} + n\pi$

$x = \frac{\pi}{8} + n\pi$ $\quad x = \frac{5\pi}{8} + n\pi$

$x = \frac{\pi}{8}, \frac{5\pi}{8}, \frac{9\pi}{8}, \frac{13\pi}{8}$

$x = \frac{\pi}{8} + \pi$

$x = \frac{\pi}{8} + \frac{8\pi}{8}$

$x = \frac{9\pi}{8}$ ✓

$x = \frac{5\pi}{8} + \pi$

$x = \frac{5\pi}{8} + \frac{8\pi}{8}$

$x = \frac{13\pi}{8}$ ✓

4. $\cos\frac{x}{2} = \frac{\sqrt{3}}{2}$

$\cos\frac{x}{2} = \frac{\pi}{6}, \frac{11\pi}{6}$ &

$\frac{x}{2} = \frac{\pi}{6} + 2n\pi$

$x = \frac{2\pi}{6} + 4n\pi$

$x = \frac{\pi}{3} + 4n\pi$

$\frac{x}{2} = \frac{11\pi}{6} + 2n\pi$

$x = \frac{22\pi}{6} + 4n\pi$

$x = \frac{10\pi}{3} + 4n\pi$ x

$x = \frac{11}{3} + 4\pi$

$x = \frac{\pi}{3} + \frac{12\pi}{3}$

$x = \frac{13\pi}{3}$ x

$x = \frac{\pi}{3}$

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.

Solve Higher-Degree Trigonometric Equations

Solve the equation on the interval $[0, 2\pi)$.

5. $2\sin^2 x + 5\sin x + 3 = 0$

$2u^2 + 5u + 3$

$(2u + 3)(u + 1)$

$2\sin x + 3 = 0$

$2\sin x = -3$

$\sin x = \frac{-3}{2}$ x

not on unit circle

$\sin x + 1 = 0$

$\sin x = -1$

$x = \frac{3\pi}{2}$

Use Identities to Solve Trigonometric Equations

Solve on the interval $[0, 2\pi)$.

6. $\sin 2x = \cos x$

$2\sin x \cdot \cos x = \cos x$

$2\sin x \cdot \cos x - \cos x = 0$

$\cos x(2\sin x - 1)$

$\cos x = 0$

$x = \frac{\pi}{2}, \frac{3\pi}{2}$

$2\sin x - 1 = 0$

$2\sin x = 1$

$\sin x = \frac{1}{2}$

$x = \frac{\pi}{6}, \frac{5\pi}{6}$

$x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$

7. $\sin x = \cos x$

$\frac{\sin x}{\cos x} = 1$

$\tan x = 1$

$x = \frac{\pi}{4}, \frac{5\pi}{4}$

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.

Use Inverse Functions to Express Solutions to Trigonometric Equations

8. Solve on the interval $[0, 2\pi)$. $4\cos^2 x - 7\cos x + 3 = 0$

a b c

$$\frac{7 \pm \sqrt{49 - 4(4)(3)}}{8}$$

$$\frac{7 \pm \sqrt{1}}{8}$$

+ −

$\frac{7+1}{8}$ $\frac{7-1}{8}$

$\cos x = 1$ $\cos x = \frac{6}{8}$

$x = 0$ $x = \cos^{-1}\left(\frac{6}{8}\right)$

$x = .72$ & $2\pi - .72$

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.